

Bond Pricing & Interest-Rate Models

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1. Objective

Price zero-coupon bonds under short-rate models by solving the bond-pricing PDE numerically, and test the numerics against an analytical solution (Vasicek) and against a mock market yield curve (Hull-White). The folder contains two notebooks, `vasicek.ipynb` and `hull_white.ipynb`.

2. Theory

The short rate r_t follows a mean-reverting process and the bond price $p(t, r)$ satisfies the Feynman-Kac pricing PDE (with $\tau = T - t$):

$$dp/d\tau = a(r) \quad dp/dr + 1/2 b^2 d^2p/dr^2 - r p, \quad p(\tau=0) = 1$$

- **Vasicek:** $dr = \alpha(\mu - r)dt + \sigma dW$, with constant long-run mean μ . A closed-form bond price exists, $P = A(T) \exp(-B(T)r_0)$, with $B = (1 - e^{-\alpha T})/\alpha$.
- **Hull-White:** same dynamics, but μ becomes a time-dependent function $\mu(t)$ chosen so the model reproduces today's observed yield curve exactly.

3. Vasicek notebook

- **Calibration:** a synthetic market yield curve $y(T) = 0.02 + 0.015(1 - e^{-0.5T})$ is fitted by least squares over $(\alpha, \mu, \sigma, r_0)$ using bounded L-BFGS (scipy). The notebook notes that Vasicek can only approximate the market curve, never match it exactly, since μ is constant.
- **PDE solve:** $\alpha = 0.5$, $\mu = 0.05$, $\sigma = 0.01$, $T = 1$ on r in $[-0.4, 0.4]$, using Robin-type boundary conditions $dp/dr + B(\tau)p = 0$ at both ends and the initial condition $p(0, r) = 1$ (a single Chebyshev T_0 mode).
- **Validation:** the numerical surface is compared with the analytical Vasicek price, and a heatmap of $\log_{10}$ absolute error over (τ, r) is produced, along with the Chebyshev-coefficient decay plot as a spectral-convergence check.

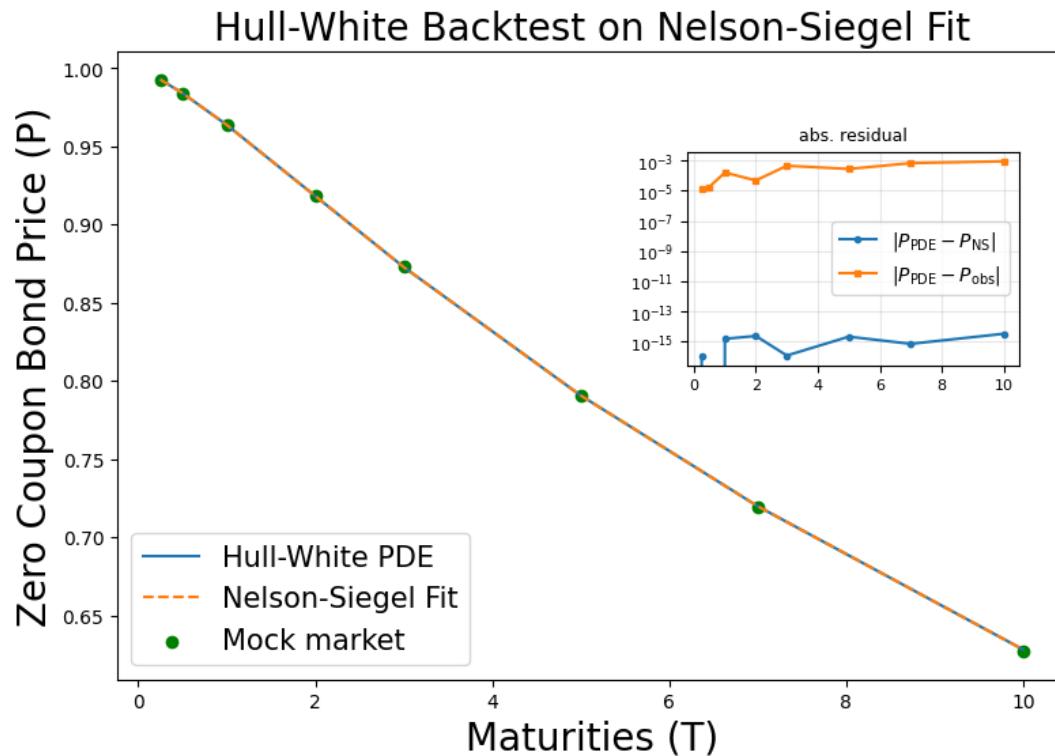
Shared numerical engine. Every PDE notebook uses the same spectral method-of-lines machinery: the solution is expanded in Chebyshev polynomials ($n = 50$ modes; the differentiation matrix is loaded from `diffmatrix100.txt`), the PDE is enforced at Gauss-Lobatto collocation nodes, and boundary conditions replace rows of the collocation system. Time is then advanced with an 8-stage singly diagonally implicit Runge-Kutta (SDIRK/DIRK) scheme at a step of 10^{-3} .

4. Hull-White notebook

This notebook closes the loop between market data, calibration and PDE pricing:

- **Mock market:** a Nelson-Siegel curve ($b_0 = 0.045$, $b_1 = -0.02$, $b_2 = 0.03$, $\lambda = 1.5$) is sampled at 11 maturities (3 months to 30 years) with small Gaussian noise (2 bp).

- **Nelson-Siegel fit:** a two-level fit - for fixed lambda the level/slope/curvature (b_0, b_1, b_2) solve an exact linear least-squares problem, and a bounded 1-D search finds lambda. This is deterministic and avoids the local minima of a 4-parameter nonlinear fit.
- **Drift function:** the fitted curve yields the instantaneous forward $f(0, t)$ and its derivative, giving $\mu(t) = f + f'/\alpha + \sigma^2/(2\alpha^2) (1 - e^{-2\alpha t})$, and $r_0 = f(0, 0)$.
- **PDE back-test:** the Hull-White PDE ($\alpha = 0.5, \sigma = 0.01$) is solved for maturities up to the 10-year point and the PDE bond price at $r = r_0$ is compared with the Nelson-Siegel price and the mock market prices, with an inset showing absolute residuals on a log scale.



Hull-White back-test: PDE price vs Nelson-Siegel fit vs mock market (hull_white.png).

5. Files

File	Description
vasicek.ipynb	Vasicek calibration, PDE solver, error heatmap vs analytical price
hull_white.ipynb	Nelson-Siegel fit, Hull-White $\mu(t)$, PDE solve and back-test
hull_white.png	Back-test figure
diffmatrix100.txt	100x100 Chebyshev differentiation matrix used by the solver